

SWAPNIL SALGAR

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OBJECTIVE

AI/ML Engineer with 6 years of experience in Data Science, Machine Learning, and Generative AI, specializing in building production-grade intelligent systems and LLM-powered applications and deploying scalable AI/ML solutions. Proficient in Python, SQL, and various machine learning frameworks, specializing in Generative AI, Agentic AI and LLM fine-tuning. Demonstrated success in building end-to-end systems and collaborating with cross-functional teams to deliver impactful AI-driven solutions. Expertise in classical ML models, deep learning architectures, and advanced NLP techniques.

PROFILE SUMMARY

- Expertise in Generative AI including LLM fine-tuning (LoRA, QLoRA), RAG-based systems, prompt engineering, and Agentic AI development using LangGraph and LangChain for autonomous reasoning and multi-agent workflows
- Specialized in Agentic AI development using LangGraph and LangChain to build autonomous agents with reasoning, planning, memory systems, and tool-calling capabilities for complex multi-step workflows
- Proficient in working with leading LLMs: OpenAI GPT, Llama, BERT, Anthropic, Mistral, and Hugging Face models for NLP, content generation, and conversational AI
- Skilled in GenAI techniques and frameworks: LangChain, LangGraph, LlamaIndex, Hugging Face Transformers, vector databases (Pinecone, FAISS, ChromaDB), Databricks, Streamlit, and Gradio
- Strong foundation in classical ML (Random Forest, XGBoost, SVM) and deep learning (TensorFlow, Keras, OpenCV) with experience in CNNs, LSTMs, and Siamese networks for CV, NLP, and fraud detection
- Proficient in Python, SQL, and backend integration using Flask/FastAPI for deploying scalable AI solutions
- Proven ability to collaborate with cross-functional teams to deliver AI-powered solutions that drive business value and improve decision-making

EDUCATION

- MBA (Part-Time, IT) – BAMU University | 2021–2024 | CGPA: 8.00
- M.Tech (Thermal Engg) – VJTI, Mumbai | 2018–2020 | CGPA: 8.88
- B.E. (Mechanical Engg) – Solapur University | 2012–2016 | 75.25%

EXPERIENCE

Brillio Technologies (Senior Data Scientist)

May- 2025 to Current

1. Architected AI-driven code conversion system:

Achieving 60-70% reduction in manual effort for legacy SAS ETL script migration to Python/PySpark. Executed token-aware chunking strategy for LLM-based semantic translation of large scripts within API constraints. Developed custom SAS parser to produce structured AST representations for complex constructs such as macros and PROC SQL.

Designed multi-agent orchestration system using LangGraph for script analysis, conversion, validation, and documentation management.

- Created dependency resolution engine with NetworkX to ensure proper execution order and dataset lineage.
- Generated automated visualization pipelines producing Mermaid diagrams illustrating process flow and data dependencies.
- Facilitated scalable modernization of enterprise analytics systems while maintaining business logic integrity, significantly reducing costs for Fortune clients.

2. AI-Powered R Shiny Application Analyzer

Large R Shiny applications often consist of multiple inter-dependent R scripts (ui.R, server.R, helper modules), making them difficult to understand, maintain, and document.

Bajaj Finance Limited (Senior Data Scientist)

June-2022 to May-2025

- Spearheaded design and implementation of AI/ML solutions to boost home loan engagement and conversion.
- Engineered predictive models employing decile scoring for effective marketing strategies.
- Developed GenAI-driven tools, including an RAG-based chatbot for tailored loan recommendations.
- Established comprehensive data pipelines integrating LLMs like OpenAI and LangChain to enhance performance.
- Partnered closely with business teams to improve operational efficiency and influence sales outcomes.
- Elevated customer interaction quality, prioritized leads, and increased loan conversions through strategic initiatives.
- Orchestrated Smart Financial Assistant project utilizing RAG-based chatbot for instant eligibility assessments.
- Executed targeted marketing solutions leveraging predictive analytics to identify high-potential home loan customers.

Thermax Ltd - Data Analyst (Process Engineer)

Jul-2021 to Mar-2022

- Forecasted daily fuel consumption for industrial boilers using machine learning techniques.
- Analyzed historical data, including steam load, weather, and efficiency metrics.
- Developed regression model to predict future fuel needs accurately.
- Optimized procurement strategies to minimize operational costs effectively.
- Ensured uninterrupted operations through precise fuel consumption estimates.
- Enhanced planning accuracy by leveraging data-driven decision-making processes.
- Utilized tools such as Python, pandas, numpy, and scikit-learn for model development.

Infinite Tech Solutions Pvt Ltd (Jr. Data Scientist)

Jan-2020 to July-2021

- Designed end-to-end segmentation pipeline using RFM scoring to quantify customer engagement.
- Performed feature engineering on raw transaction data to derive and normalize RFM metrics.
- Implemented K-Means clustering, optimizing cluster count with Elbow Method and Silhouette Score.
- Visualized customer segments through scatter plots and bar charts for business insights.
- Delivered actionable marketing insights that enhanced campaigns for high-value and at-risk segments.
- Achieved over 20% increase in campaign ROI and customer retention through targeted outreach.
- Preprocessed signature datasets using data augmentation techniques to improve model generalization.
- Developed custom Siamese Neural Network for automated signature verification, enhancing fraud detection efficiency.

SKILL SET

- Programming: Python, SQL
- Classical ML: Logistic Regression, Random Forest, XGBoost, SVM
- Deep Learning: CNNs, LSTMs, Siamese networks, TensorFlow, Keras, OpenCV
- Generative AI: LLMs, LoRA/QLoRA fine-tuning, RAG, prompt engineering, GANs, VAEs, Diffusion Models
- Agentic AI: LangGraph, LangChain, multi-agent workflows, tool-calling, memory & state management
- LLMs & NLP: OpenAI GPT, Llama, BERT, Mistral, Anthropic, Hugging Face Transformers, text classification, NER, summarization, Q&A
- Vector & Knowledge Systems: Pinecone, FAISS, ChromaDB, LlamaIndex, retrieval pipelines
- Backend & APIs: FastAPI, Flask, REST APIs, integration of ML/LLM services
- MLOps & Data: Databricks, model deployment, data pipelines, monitoring
- Apps & Prototyping: Streamlit, Gradio, PoC to production transition